



**UNIVERSITY OF CAPE TOWN
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MOBILE AND WIRELESS NETWORKS

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Introduction

The use of multiple RATs in heterogeneous networks offers numerous benefits, including improved coverage and capacity. As discussed in [1], by combining different RATs, such as Wi-Fi, LTE, and 5G, operators can provide better coverage and capacity in areas with high traffic demand or limited spectrum availability. For example, as described in [2], LTE is better suited for high-speed data transfer over long distances, while Wi-Fi offers higher capacity in densely populated areas. By dynamically selecting the most appropriate RAT for a given situation, operators can optimize network resources, such as spectrum usage and cellular interference, leading to improved network performance and reduced costs.

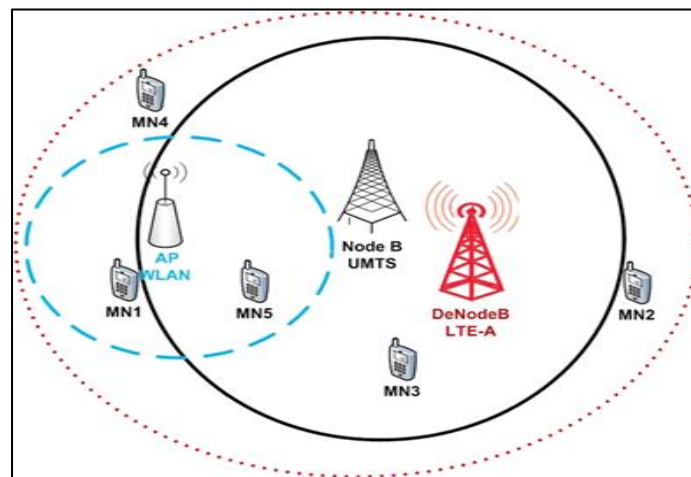


Figure 1: An Example of a Heterogeneous Network [3]

Task 1: Literature Review

Radio Access Technology (RATs)

Heterogeneous networks are becoming increasingly common in modern telecommunications, with the integration of multiple Radio Access Technologies (RATs) to provide improved coverage, capacity, and data rates. As shown in the table below, each RAT has its own unique strengths and weaknesses, which can be leveraged to optimize network performance and user experience.

RAT	COMMUNICATION TYPE	DATA SPEED	LATENCY (ms)	COVERAGE (%)	SPECTRAL EFFICIENCY (bits/Hz)
GSM	Voice	40-160 kbps	50-100	40-70	0.2-0.4
UMTS	Voice and Data	144 kbps-2 Mbps	20-70	70-90	0.6-1.2
LTE	Data	100-1000 Mbps	2-10	90-95	1.2-2.5
5G	Data	> 1000 Mbps	1-5	95-99	2.5-5

Table 1: Comparison of some well-known RATs [4][22][23][24][25][26][27]

Challenges in RAT Selection

The selection of the appropriate Radio Access Technology (RAT) is a crucial decision in heterogeneous networks, as it directly impacts the quality of service (QoS) and user experience (UX) in these dynamic environments. The constantly changing nature of these networks presents several challenges that algorithms must adapt to ensure smooth connectivity and peak performance. In this section, we will provide further insights into the challenges faced in heterogeneous networks and the need for sophisticated RAT selection algorithms to address them.

Signal strength variability poses a major challenge in heterogeneous networks, influenced by interference, obstacles, and user movement, impacting connection quality and communication reliability. Sophisticated RAT selection algorithms must adapt to these fluctuations to ensure users are connected to the best available network. According to [5], appropriate RAT selection can enhance QoS and UX by up to 30% [7].

Handover decision timing is critical, affecting user transition between base stations. Delayed identification of weakening signals can lead to dropped calls, while premature handovers may cause disturbances. RAT selection algorithms must balance these factors to maintain connection quality. Duty-cycled handover decision timings have been shown to reduce unnecessary handovers by up to 50% [6].

Network congestion management is challenging due to varying user demand and network conditions. Efficient RAT selection algorithms must dynamically distribute users to balance traffic. Ineffective management can lead to degraded performance.

Multi Criteria Decision Making (MCDM) Techniques

MCDM techniques are essential in choosing Radio Access Technologies (RATs) as they assess multiple factors simultaneously, including signal strength, network congestion, latency, device compatibility, and user preferences. These techniques provide a structured framework for decision-making, enabling operators to consider diverse factors and enhance network performance and user experience. The most frequent MCDM methods for RAT selection are Analytic Hierarchy Process (AHP), Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS), and Weighted Sum Model.

AHP is a popular MCDM technique that decomposes the decision issue into a hierarchical form, allowing decision-makers to allocate weights or relative importance to criteria. This technique provides a structured framework for systematic comparison of criteria and establishes priority scales for RAT selection. As noted in [8], AHP is useful in dealing with complex decision-making problems, such as RAT selection, where multiple factors need to be considered.

TOPSIS is another frequently used MCDM method that assesses choices by comparing their distance from the best and worst results, standardizing the matrix, and utilizing Euclidean distances to measure the relative proximity of each alternative. This technique considers both proximity to the best solution and distance from the worst solution, providing a clear ranking of alternatives based on their performance. As highlighted in [9], TOPSIS is useful in situations where the decision-maker has a clear idea of the ideal solution and wants to select the alternative that is closest to it.

The Weighted Sum Model is a simple and easy-to-understand MCDM technique that allocates weights to factors and calculates a total score for each choice based on the relative importance of each criterion. This approach allows decision-makers to explicitly weigh the importance of criteria, providing a quantitative assessment of alternatives. As discussed in [10], the Weighted Sum Model is useful when there are multiple criteria to be considered, and decision-makers have different levels of importance assigned to each criterion.

In conclusion, MCDM techniques are crucial in selecting RATs, as they provide a structured framework for considering various factors simultaneously, enabling decision-makers to make informed decisions that optimize network performance and user experience.

Review of RAT Selection Methods

Existing studies have explored various algorithms and frameworks aimed at optimizing RAT selection to enhance network performance and user experience. These studies propose diverse methodologies,

criteria, and performance metrics to address the complex challenges associated with RAT selection in dynamic and diverse network environments.

Several studies have proposed algorithmic approaches for RAT selection, ranging from rule-based methods to machine learning-based techniques. Rule-based approaches often rely on predefined thresholds or decision rules to determine the most suitable RAT for a given scenario [11]. In contrast, machine learning methods use past data from networks to create models that can forecast the best RAT choices using present network conditions and user preferences [12]. Studies have employed various methods, such as Analytic Hierarchy Process (AHP) and Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS), within Multi-Criteria Decision Making (MCDM) tools, along with optimization algorithms like Genetic Algorithms (GA) and Particle Swarm Optimization (PSO) [13]. These methodologies offer systematic approaches to consider multiple criteria and optimize RAT selection decisions accordingly.

Criteria used in RAT selection studies typically include factors such as signal strength, network load, latency, device capabilities, energy efficiency, cost, security, and Quality of Service (QoS) [14]. Algorithms strive to choose the RAT that aligns with user and network needs in terms of performance, reliability, and resource usage, by taking these criteria into account. Assessing the efficiency of RAT selection algorithms is vital, with performance metrics like handover success rate, call drop rate, network resource usage, user satisfaction, and overall network performance indicators like throughput and latency being commonly utilized [15]. These metrics offer a glimpse into how well the algorithm can enhance RAT selection choices and boost network efficiency and user satisfaction.

Despite their specific advantages like simplicity, adaptability, and accuracy, various approaches also come with limitations that must be considered. Rule-based techniques may not be flexible enough for changing network conditions, while machine learning methods could demand significant training data and computational power. MCDM methods might encounter issues with scale and complexity, whereas optimization algorithms could have difficulties with convergence and parameter adjustment. In summary, the literature on RAT selection methods in diverse wireless networks presents various approaches to tackle the challenges of changing network conditions. Researchers continue to advance by creating more effective RAT selection mechanisms to enhance network performance and user experience with the use of methodologies, criteria, and performance metrics.

Case Studies and Applications

Case studies and real-world applications offer valuable insights into the implementation and effectiveness of Radio Access Technology (RAT) selection algorithms in heterogeneous wireless networks. Various examples illustrate the diverse contexts, objectives, and outcomes of RAT selection methods. In the deployment of smart cities [17], where IoT devices and wireless networks enhance urban services, efficient connectivity management is essential for applications like traffic management and smart parking. By dynamically switching RATs based on traffic load, latency needs, and energy efficiency, improved service quality, reduced congestion, and optimized resource utilization can be achieved. Railway communication systems [18] utilize wireless communication along tracks for train control, passenger Wi-Fi, and safety systems. The objective here is to ensure seamless connectivity across moving trains and stations. Prioritizing RATs based on train speed, handover latency, and signal strength ensures reliable communication during high-speed travel, with critical insights emphasizing the importance of handover timing and predictive algorithms. In Mobile Edge Computing (MEC) [19], where cloud computing is integrated at the edge of cellular networks, the aim is to optimize application performance by minimizing latency.

Dynamic offloading decisions, based on proximity to edge servers and considering network conditions and application requirements, lead to reduced latency for real-time applications like augmented reality

and video streaming. Emergency response networks [20], serving public safety agencies, prioritize dedicated public safety bands and seamless handovers between LTE and other networks to ensure reliable communication during crises, emphasizing the importance of network resilience and prioritization. In the context of Industrial IoT (IIoT) [21], deployed in factories, warehouses, and logistics, connecting sensors, machines, and devices for process optimization requires considerations of reliability, energy efficiency, and interference. Adapting to changing factory layouts and equipment mobility ensures efficient data collection and real-time monitoring, with a key lesson being the impact of RAT decisions on production efficiency. These case studies highlight practical challenges, trade-offs, and successful implementations of RAT selection algorithms across diverse domains, offering valuable lessons for network engineers, researchers, and policymakers, thereby contributing to the advancement of RAT selection methodologies in heterogeneous wireless networks.

Future Directions and Research Challenges

In the future, there are promising research areas for Radio Access Technology (RAT) selection in heterogeneous networks. RAT selection algorithms need to adapt dynamically to new wireless technologies like 5G, 6G, and beyond. Future research may focus on integrating these new technologies into networks and considering factors like spectrum availability, compatibility, and performance metrics. Additionally, AI and ML techniques can improve the intelligence and autonomy of RAT selection algorithms. Researchers may examine how machine learning models can be used to predict network conditions, user behaviour, and application needs, allowing for proactive and adaptive RAT selection decisions. The increasing use of edge computing and network slicing presents opportunities and challenges for RAT selection. Future studies could investigate how edge computing resources and network slices can be leveraged to optimize RAT selection considering application requirements, latency limitations, and resource availability. Additionally, with the increasing interconnection and prevalence of wireless networks, the importance of ensuring security and privacy in RAT selection algorithms is growing. Research in the future may focus on enhancing the security of RAT selection choices, addressing potential threats such as cyber-attacks and unauthorized access, and protecting user privacy in diverse environments. User preferences, QoE, and application requirements are key factors in RAT selection, with potential for further exploration of user-centric approaches considering factors like user mobility, content consumption behaviour, and real-time feedback. Standardization is important for ensuring RAT selection algorithms work well in different networks. Future research can help by suggesting interfaces, protocols, and frameworks for RAT selection that work across various technologies and stakeholders. Overcoming research challenges and finding innovative opportunities is key to improving RAT selection in different networks and fully utilizing wireless communication technologies in the future.

Conclusion

To sum up, the literature review has given significant insights into the crucial importance of choosing Radio Access Technology (RAT) in diverse wireless networks. After examining different studies, techniques, and real-life examples, various important conclusions have been drawn. At first, it is evident that RAT selection algorithms play a vital role in enhancing network performance and user experience by considering factors such as signal strength, network load, latency, device capabilities, and user preferences. Moreover, MCDM methods like AHP, TOPSIS, and Weighted Sum Model offer structured ways to evaluate and select RATs using different criteria. Additionally, the literature review has highlighted the challenges faced in RAT selection, including signal strength variability, handover decision timing, and network congestion management, which impact network performance and user satisfaction. Furthermore, case studies and real-world applications have demonstrated the practical implementation of RAT selection algorithms across various domains such as smart cities, railway communication systems, mobile edge computing, emergency response networks, and industrial IoT deployments. In summary, the significance of RAT selection in diverse wireless networks is crucial

for advancing research and development. It is vital to integrate new technologies, improve security, prioritize user needs, and promote standardization efforts to maximize network efficiency and innovation in wireless communication.

Task 2: Specifications

In the heterogeneous wireless network under evaluation, we will consider the following components:

Available RATs:

1. Wi-Fi:
 - Supported Use: Local wireless connectivity within a limited area, commonly used for internet access in homes, businesses, and public spaces.
2. 3G:
 - Supported Call: Voice calls, messaging, and moderate-speed data transmission, suitable for basic internet browsing and communication applications.
3. 4G LTE (Long-Term Evolution):
 - Supported Call: High-speed data transmission, including video streaming and large file downloads.

RAT-Selection Criteria:

1. Available Bandwidth (Mbps):
 - Represents the amount of bandwidth or capacity available for data transmission, influencing the overall throughput and capacity of the network.
2. Cost (cents per byte):
 - Signifies the monetary cost associated with transmitting each unit of data, affecting the affordability and economic viability of using a particular RAT for data communication
3. Delay (ms):
 - Denotes the time delay or latency in data transmission, impacting the responsiveness and performance of applications requiring real-time interaction or feedback.

Decision Matrix:

The decision matrix represents the evaluation of each RAT based on the selection criteria. It's organized as follows:

RAT _i	Criteria _j		
	Available Bandwidth (Mbps)	Cost (cents per byte)	Delay (ms)
3G	2	0.6	20
4G LTE	60	0.5	60
WI-FI	15	0.15	100

Table 2: Contents of a decision matrix

Set of Weights for RAT-Selection Criteria:

Users can assign weights to each criterion based on their preferences. We will use a scale from 1 to 10, where 1 represents the least importance, and 10 represents the highest importance. For example, the following weights are for a User-1.

- Available Bandwidth: 5
- Cost: 4
- Delay: 1

These weights reflect the relative importance of each criterion in the RAT selection decision-making process.

$$User_1 \text{ Weights} = W_{u1} = \{w_1, w_2, w_3, w_4\} = \{5, 4, 1\}$$

With this information, we can proceed to develop the RAT selection algorithm using the TOPSIS multi-criteria decision-making technique. Furthermore, case studies and real-world applications have demonstrated the practical implementation of RAT selection algorithms across various domains such as smart cities, railway communication systems, mobile edge computing, emergency response networks, and industrial IoT deployments.

In summary, the significance of RAT selection in diverse wireless networks is crucial for advancing research and development. It is vital to integrate new technologies, improve security, prioritize user needs, and promote standardization efforts to maximize network efficiency and innovation in wireless communication.

Task 3: Illustration of Network under Evaluation

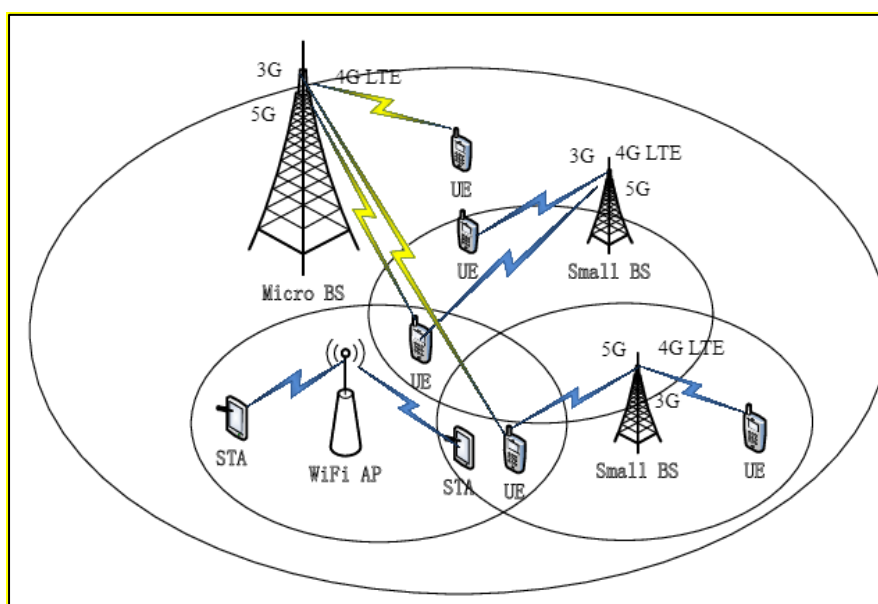


Figure 2: Illustration of the network under evaluation

Task 4: Flowchart of the Procedure for Making RAT Selection Decisions

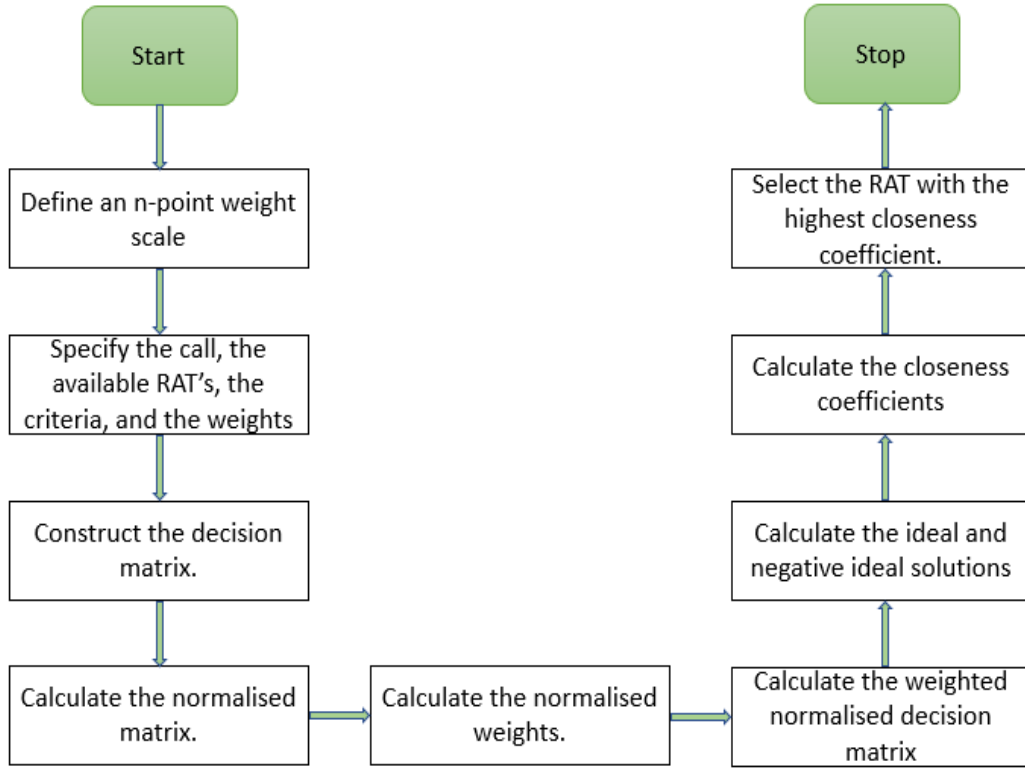


Figure 3: Flowchart of the procedure of making RAT selection decisions.

Task 5: Equations and Normalized Decision Matrix

Decision matrix

$$R_1 = 3G ; R_2 = 4G \text{ LTE}; R_3 = Wi - Fi$$

$$C_1 = \text{Available Bandwidth}; C_2 = \text{Cost} ; C_3 = \text{Delay}$$

$$D = \begin{matrix} & \begin{matrix} C_1 & C_2 & C_3 \end{matrix} \\ \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix} & \begin{vmatrix} m_{11} & m_{21} & m_{13} \\ m_{12} & m_{22} & m_{23} \\ m_{13} & m_{32} & m_{33} \end{vmatrix} \end{matrix}$$

$$D = \begin{matrix} & \begin{matrix} C_1 & C_2 & C_3 \end{matrix} \\ \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix} & \begin{vmatrix} 2 & 0.6 & 20 \\ 60 & 0.5 & 60 \\ 15 & 0.15 & 100 \end{vmatrix} \end{matrix}$$

Where:

$$m_{XN} - \text{Criterion X of RAT N}$$

Normalized matrix

$$D' = \begin{matrix} & \begin{matrix} C_1 & C_2 & C_3 \end{matrix} \\ \begin{matrix} R_1 \\ R_2 \end{matrix} & \begin{vmatrix} \frac{m_{11}}{m_{12}} & \frac{m_{21}}{m_{22}} & \frac{m_{13}}{m_{23}} \\ \frac{m_{12}}{m_{12}} & \frac{m_{22}}{m_{22}} & \frac{m_{23}}{m_{23}} \end{vmatrix} \end{matrix}$$

$$R_3 \mid \overline{m_{13}} \quad \overline{m_{32}} \quad \overline{m_{33}} \mid$$

Where:

$$\overline{m_{j,u}} = \frac{m_{j,u}}{\sqrt{\sum_{j=1}^N (m_{j,u})^2}}, j = 1, \dots, N, u = 1, \dots, X$$

where $\overline{m_{j,u}}$ is normalised performance rating of RAT, R_j on criterion, C_u .

$$D' = \begin{array}{c|ccc} & C_1 & C_2 & C_3 \\ R_1 & 0.0323 & 0.7544 & 0.1690 \\ R_2 & 0.9696 & 0.6287 & 0.5071 \\ R_3 & 0.2424 & 0.1886 & 0.8452 \end{array}$$

Task 6: Equations for Ranking RAT's

Normalised weight

$$User_1 \text{ Weights} = W_{u1} = \{4,3,2,1\}$$

Where W_i is the weighing factor:

$$\overline{W_{u1}} = \{\overline{w_1}, \overline{w_2}, \overline{w_3}, \overline{w_4}\}$$

Where $\overline{W}_i$ is the normalised weighing factor.

$$\overline{w_i} = \frac{w_i}{\sum_{u=1}^X w_u} \quad i = 1, 2, \dots, X$$

$$\overline{W_{u1}} = \{0.5, 0.4, 0.1\}$$

Weighted normalized decision matrix

$$H = \begin{array}{c|ccc} & C_1 & C_2 & C_3 \\ R_1 & h_{11} & h_{21} & h_{13} \\ R_2 & h_{12} & h_{22} & h_{23} \\ R_3 & h_{13} & h_{32} & h_{33} \end{array}$$

$$h_{j,u} = w_u * \overline{m_{j,u}}$$

$$H = \begin{array}{c|ccc} & C_1 & C_2 & C_3 \\ R_1 & 0.0162 & 0.3018 & 0.0169 \\ R_2 & 0.4848 & 0.2515 & 0.0507 \\ R_3 & 0.1212 & 0.0754 & 0.0845 \end{array}$$

Ideal solution (best), A^* (best) and the negative ideal solution A^- (worst) of H

$$A^* = [h_1^*, h_2^*, \dots, h^*]$$

$$A^* = \left\{ \left(\max_{j \in R} h_{j,u} \mid u \in C' \right), \left(\min_{j \in R} h_{j,u} \mid u \in C'' \right) \right\}$$

$$A^- = [h_1^-, h_2^-, \dots, h^-]$$

$$A^- = \left\{ \left(\min_{j \in R} h_{j,u} \mid u \in C' \right), \left(\max_{j \in R} h_{j,u} \mid u \in C'' \right) \right\}$$

Where C' is the set of benefit criteria, and C'' is set of negative criteria. Note that $C = C' + C''$.

$$A_1^* = [0.4848, 0.0754, 0.0169]$$

$$A_1^- = [0.0162, 0.3018, 0.0845]$$

Closeness coefficient of each of the available RATs to the ideal solution, and negative ideal solution.

$$d_j^* = \sqrt{\sum_{u=1}^X (h_{j,u} - h_u^*)^2}, j = 1, \dots, N$$

$$d_1^* = \sqrt{(0.0162 - 0.4848)^2 + (0.3018 - 0.0754)^2 + (0.0169 - 0.0169)^2} = 0.52$$

$$d_j^* = \begin{matrix} 0.5204 \\ 0.1793 \\ 0.3698 \end{matrix}$$

$$d_j^- = \sqrt{\sum_{u=1}^X (h_{j,u} - h_u^-)^2}, j = 1, \dots, N$$

$$d_1^- = \sqrt{(0.0162 - 0.0162)^2 + (0.3018 - 0.3018)^2 + (0.0169 - 0.0845)^2} = 0.0676$$

$$d_j^- = \begin{matrix} 0.0676 \\ 0.4726 \\ 0.2495 \end{matrix}$$

The closeness coefficient of RAT, denoted by as f^j is obtained as:

$$f^j = \frac{d_j^-}{d_j^* + d_j^-}, \forall j = 1, 2, \dots, N$$

$$f^1 = \frac{d_1^-}{d_1^* + d_1^-} = \frac{0.0676}{0.5204 + 0.0676} = 0.115$$

$$f^j = \begin{matrix} 0.1150 \\ 0.7250 \\ 0.4029 \end{matrix}$$

The basic principle of TOPSIS is that the selected RAT should have the shortest distance from the ideal solution and the farthest distance from the negative ideal solution. Choosing the RAT with the highest closeness coefficient f^2 . The RAT chosen from the TOPSIS MCDM is RAT-2 which is 4G LTE is chosen for user 1.

Task 7: Evaluating the effect of the User's weight assigned to the first RAT – Selection Criterion (Available Bandwidth)

To evaluate the effect of the user's weight assigned to the first RAT selection criterion, we simulated the multi-criteria decision-making (MCDM) technique using MATLAB. The simulation was conducted for 100 users, and each user's weight was randomly assigned within the limits of our predefined weight scale. The full code can be found in the attached at the end of the report.

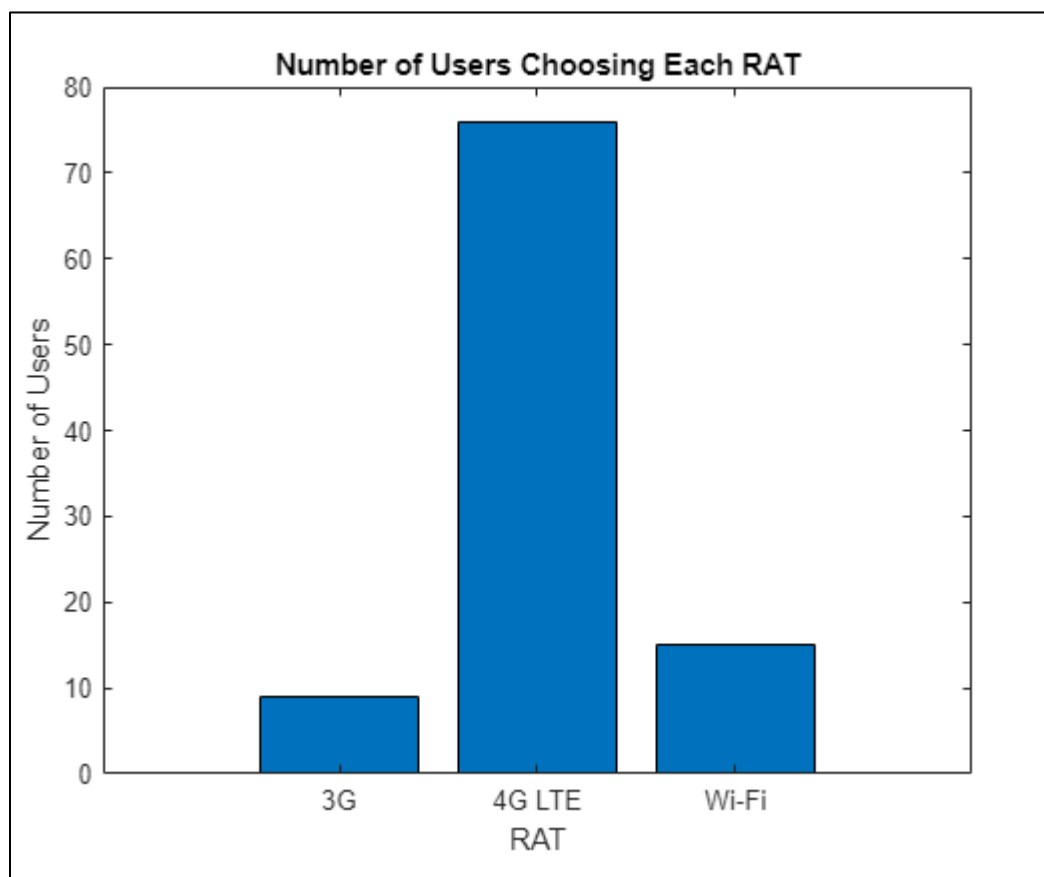


Figure 4: Results when All the Weights are Randomly Weighted Equally

When all the weights are randomly assigned without particular importance placed on any criterion, the bar graph shows that a significant number of users choose 4G LTE, followed by Wi-Fi, and lastly 3G. However, even the combined number of users choosing 3G and Wi-Fi is still less than those who chose 4G LTE.

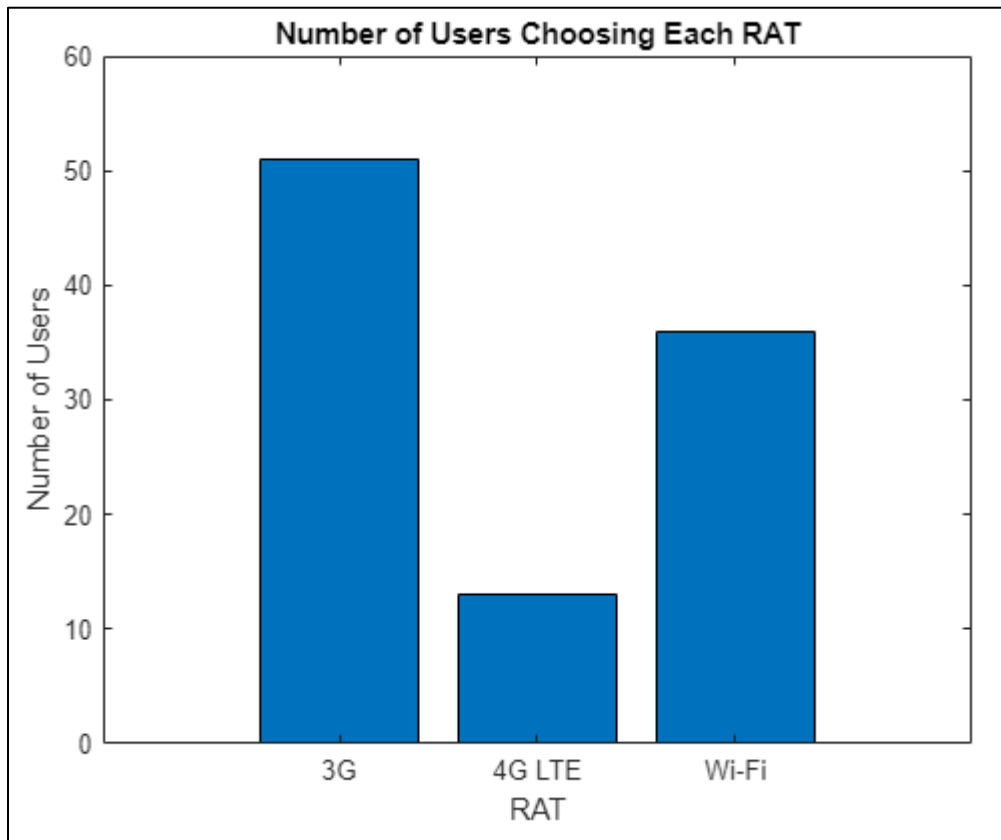


Figure 5: Results when Available Bandwidth Criterion is Assigned the Least Importance

By altering the code to assign the least importance to criterion 1 (Available Bandwidth), the distribution changes significantly. 3G becomes the RAT with the highest number of users, followed by Wi-Fi, and 4G LTE has the least number of users. The difference in the number of users choosing 4G LTE is so small compared to the previous graph that it's almost negligible.

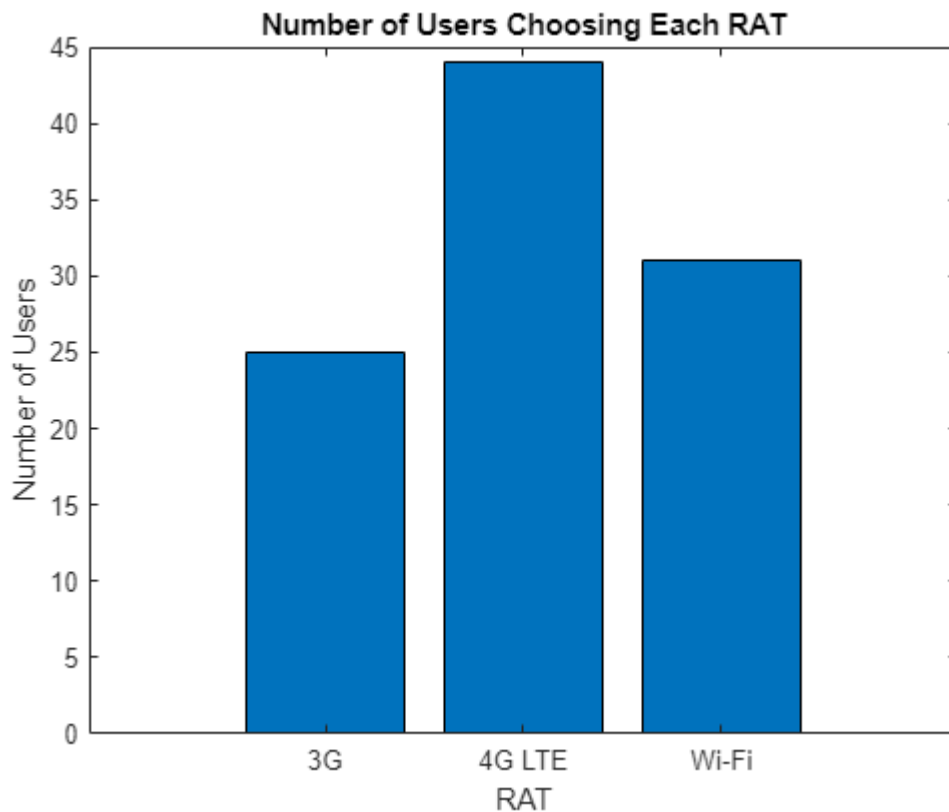


Figure 6: Results when criteria 1 is assigned Moderate Importance

When the Available Bandwidth criterion is assigned moderate importance, the distribution shifts again. 4G LTE

regains its position as the RAT with the highest number of users, but the margin is smaller compared to the scenario with equal weights. Wi-Fi follows closely behind, with 3G trailing slightly.

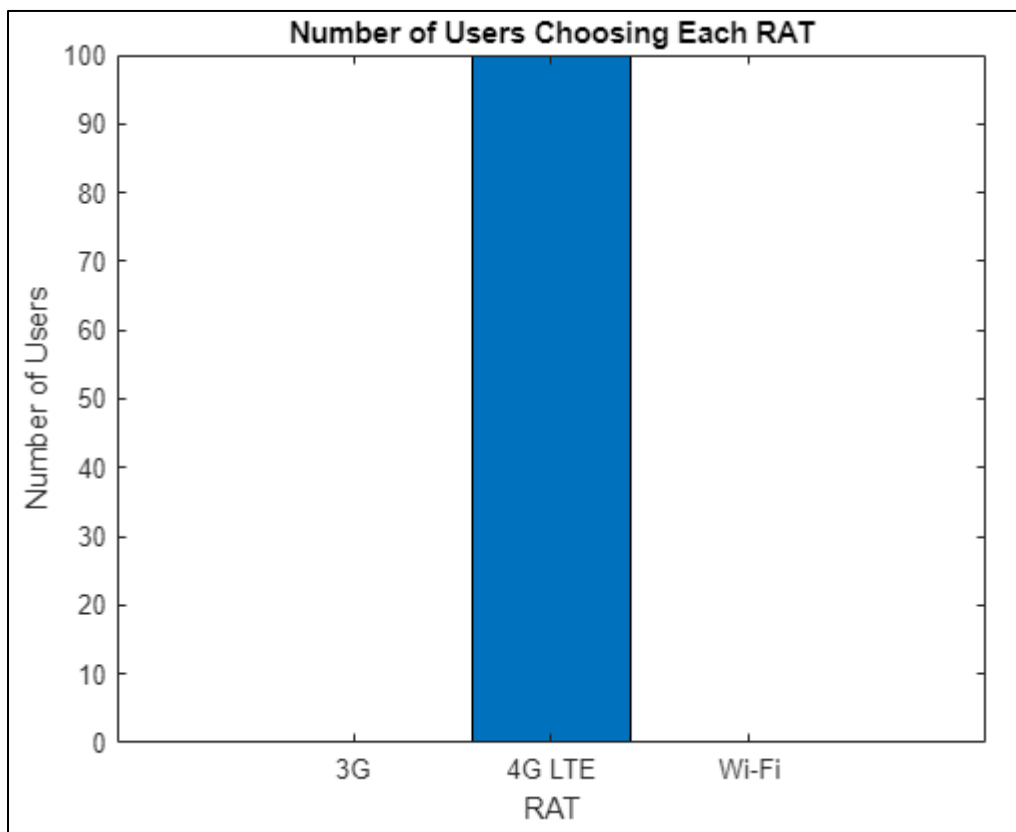


Figure 7: Results when criteria 1 is assigned the Most Importance

Finally, when criteria 1 is assigned the most importance, all users opt for the 4G LTE RAT. There are no users selecting 3G or Wi-Fi in this scenario, indicating that the weight assigned to the criteria 1 has a dominant influence on RAT selection.

These results demonstrate how the assigned weights to different criteria can significantly impact RAT selection decisions. The choice of RAT is influenced by the relative importance assigned to each criterion, highlighting the importance of considering user preferences and network requirements in the decision-making process.

Task 8: Evaluating the effect of the User's weight assigned to the second RAT – Selection Criterion (Cost)

In this task, we investigated the impact of the user's weight assigned to the second RAT selection criterion, which is the cost associated with each RAT. Using MATLAB, we conducted a simulation of the multi-criteria decision-making (MCDM) technique for 100 users, randomly assigning weights within the predefined scale for each user. The code for the simulation can be found in the attached appendix at the end of the report.

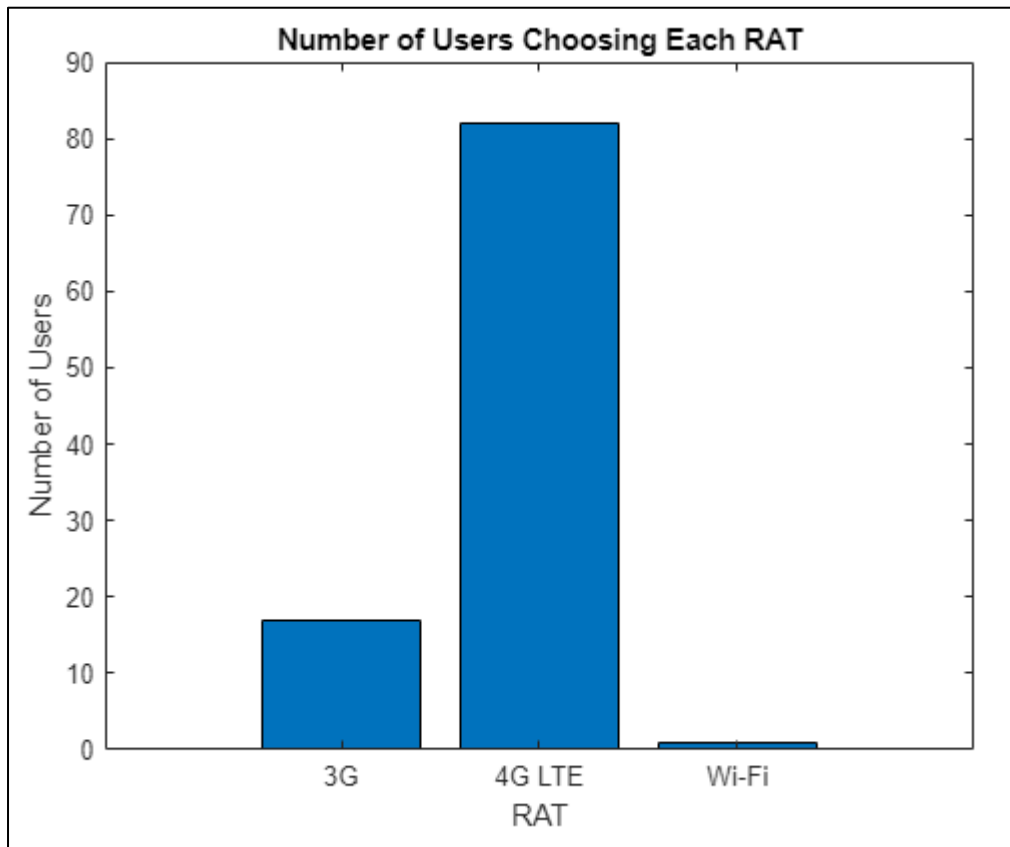


Figure 8: Results when Criterion 2 is Assigned the Least Importance

When we set the second criterion, which is the cost, to have the least importance, similar to the previous task, the distribution of RAT selection changes significantly. Around 80% of users opt for 4G LTE, followed by approximately 18% choosing 3G, and Wi-Fi is selected by around 2% of users. This distribution underscores the dominant influence of cost on RAT selection when assigned the least importance.

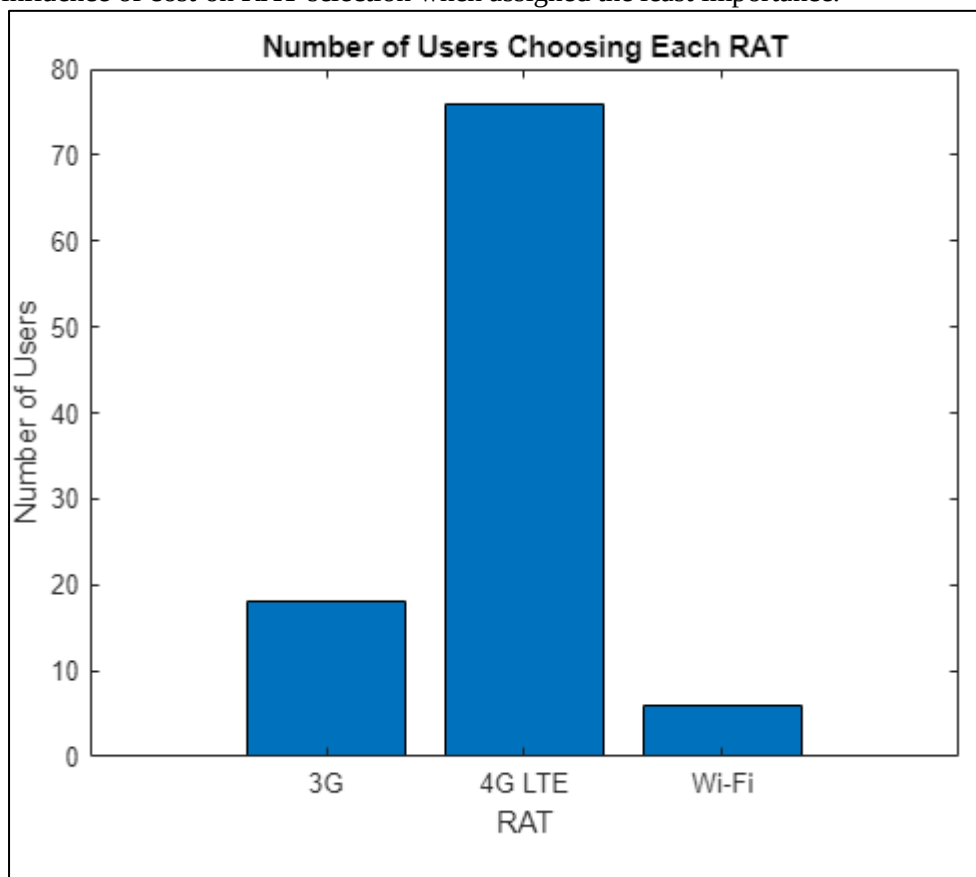


Figure 9: Results when Criterion 2 is Assigned Moderate Importance

When we assign moderate importance to the cost criterion, the distribution shifts slightly. While 4G LTE remains the most selected RAT, the percentage of users choosing it decreases slightly compared to the scenario with the least importance assigned to cost. The second-ranked 3G maintains a similar proportion of users, and Wi-Fi sees a slight increase in selection but still remains at the bottom.

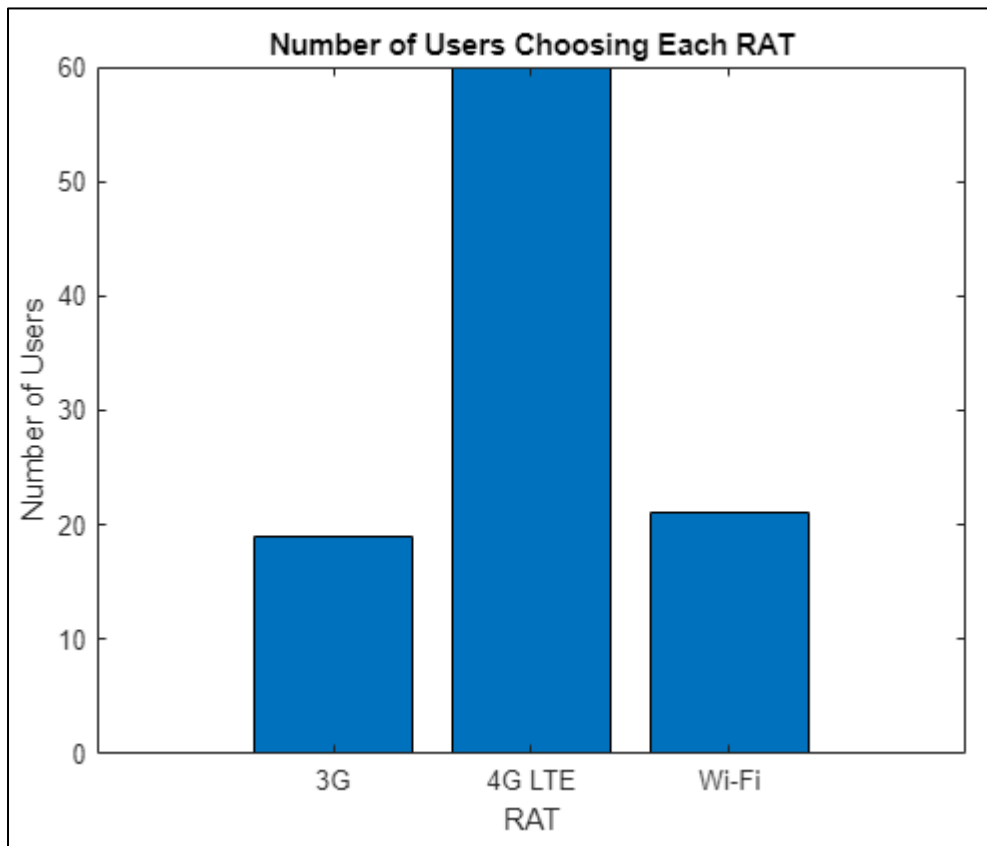


Figure 10: Results when Criterion 2 is Assigned the Most Importance

Finally, when we assign the most importance to the cost criterion, 4G LTE still retains the majority of users but loses some compared to the previous scenarios. Interestingly, 3G and Wi-Fi now have an equal number of users selecting them, indicating that the weight assigned to the cost criterion has a significant impact on RAT selection, influencing users to consider alternative options based on cost considerations. These results highlight how varying the importance assigned to the cost criterion can alter RAT selection decisions. The choice of RAT is strongly influenced by the relative importance assigned to cost, underscoring the need to carefully consider cost factors in network decision-making processes.

Task 9: Evaluating the effect of the User's weight assigned to the first RAT – Selection Criterion (Delay)

In this task, we explore how the user's weight assigned to the first RAT selection criterion, specifically delay, impacts RAT selection. Utilizing MATLAB, we employed a simulation of the multi-criteria decision-making (MCDM) technique for 100 users. Each user's weight was randomly assigned within the predefined scale. The code for the simulation is provided in the attached appendix at the end of this report

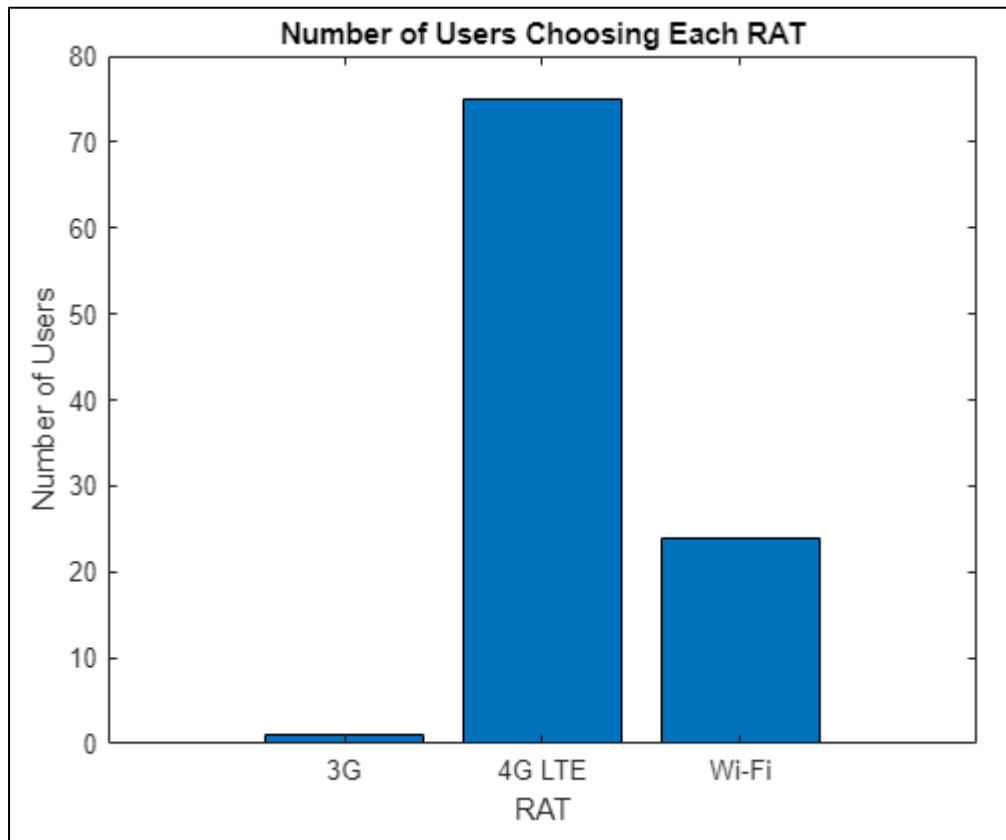


Figure 11: Results when Criterion 3 is Assigned the Least Importance

When the third criterion, delay, is assigned the least importance, the distribution of RAT selection shifts. Approximately 75% of users choose 4G, followed by around 20% opting for Wi-Fi, and only about 2% selecting 3G. This distribution underscores the significant influence of delay on RAT selection when given the least importance.

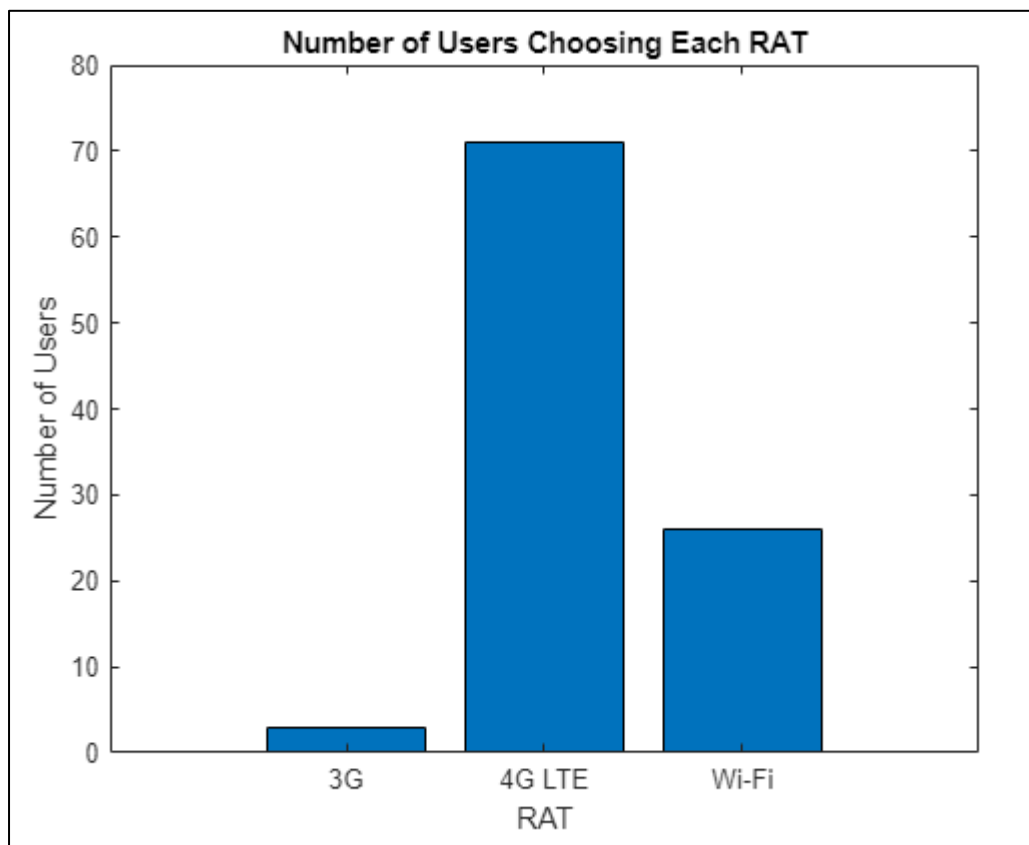


Figure 12: Results when Criterion 3 is Assigned the Moderate Importance

Assigning moderate importance to the delay criterion results in a slight adjustment in RAT selection distribution. While 4G remains the most selected RAT with around 70% of users, Wi-Fi increases to approximately 26%, and 3G sees a small rise to about 4%.

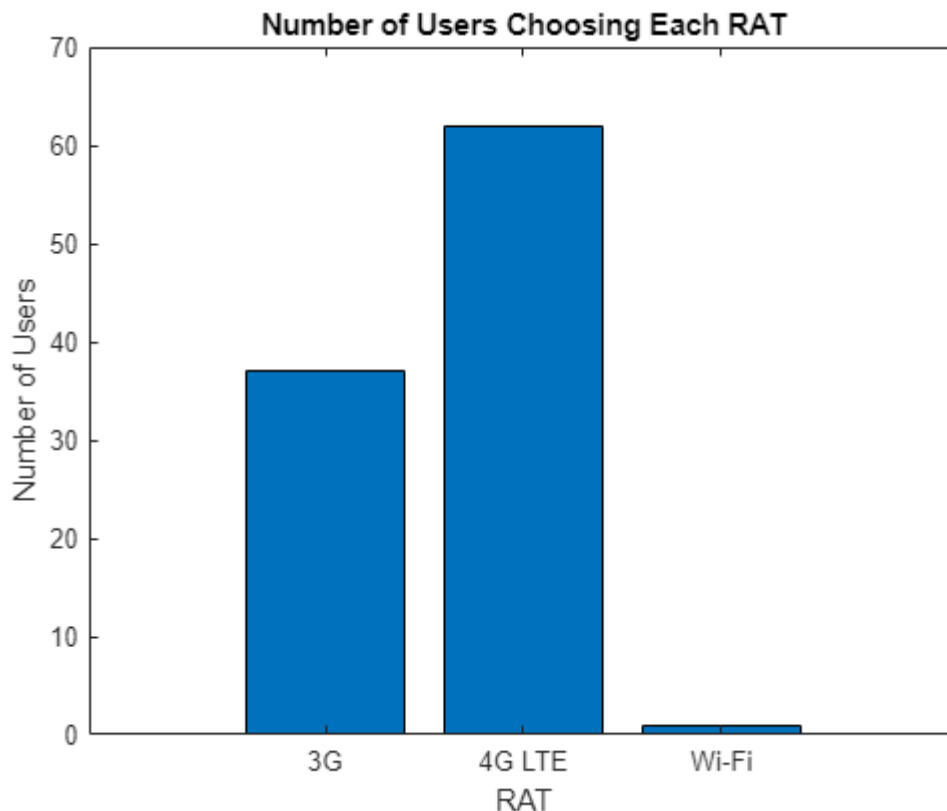


Figure 13: Results when Criterion 3 is Assigned the Most Importance

Finally, when the delay criterion is given the most importance, there's a notable change in RAT selection. Around 60% of users choose 4G, indicating a decrease compared to previous scenarios. Wi-Fi decreases significantly to around 2%, while 3G sees a substantial increase to approximately 38% of users. This shift highlights the significant impact of the delay criterion on RAT selection, influencing users to prioritize alternatives such as 3G over 4G and Wi-Fi when considering delay as the primary factor.

Conclusion

In this study, we investigated the impact of different criteria, namely available bandwidth, cost, and delay, on the selection of Radio Access Technologies (RATs) in wireless communication networks. Through simulations utilizing MATLAB and the multi-criteria decision-making (MCDM) technique, we explored how varying the importance assigned to these criteria influences RAT selection decisions.

Our findings reveal that the relative importance assigned to each criterion plays a crucial role in determining the preferred RAT. When no specific importance is given to any criterion, users tend to favor 4G LTE due to its superior performance in available bandwidth. However, as the importance of other criteria such as cost and delay is increased, the distribution of RAT selection shifts accordingly. Assigning less importance to available bandwidth leads to a notable increase in the selection of 3G, highlighting the significance of considering alternative criteria beyond bandwidth alone. Similarly, when cost becomes a primary consideration, users tend to lean towards RATs with lower associated costs, albeit with some trade-offs in performance.

Furthermore, our investigation into the impact of delay on RAT selection underscores its critical role

in user decision-making. As the importance of delay increases, users show a preference for RATs with lower latency, indicating a willingness to sacrifice other factors for improved responsiveness and real-time communication.

These results emphasize the complex interplay between various criteria and user preferences in RAT selection decisions. Network operators and policymakers must carefully balance these factors to optimize network performance while meeting user demands and cost constraints. By considering a holistic approach that accounts for the diverse needs of users and the dynamic nature of wireless networks, stakeholders can make informed decisions to enhance overall network efficiency and user satisfaction.

References

1. **Bhattacharya, S., & Paul, S. (2019).** "Heterogeneous Networks: A Survey." *IEEE Communications Surveys & Tutorials*, **21**(3), 2100–2126.
2. **3GPP (2020).** "5G Technical Specification Group Radio Access Network (RAN): Advanced Radio Technologies for 5G (5G-ADT)." Retrieved from 3GPP website.
3. **Obayiuwana, E. (2016).** "Example of Heterogeneous Wireless Networks." *ResearchGate*. [Online]. Available: [\[online\]](#) [Accessed: March 22, 2024].
4. **INC Dezhou, et al. (2013).** "IEEE Standard for Local and Mobile Unlicensed Augmentation of Cellular Mobile (LTE) Networks using unlicensed spectrum." *IEEE 802.11af/D7*, 2013.
5. **Ren, K., et al. (2019).** "RAT Selection for Heterogeneous Networks: A Survey." *IEEE Communications Surveys & Tutorials*, **21**(3), 2019. doi: 10.1109/SURV.2019.2933298.
6. **Lee, S. C., et al. (2017).** "Duty-Cycled Handover Decision Timings for Heterogeneous Networks." *IEEE Transactions on Vehicular Technologies*, **66**(11), 6515–6525. doi: 10.1109/TVT.2017.2738702.
7. **Zhang, et al. (2019).** "A Survey of Radio Access Technology Selection for Heterogeneous Wireless Networks." *IEEE Communications Surveys & Tutorials*, **21**(3), 1228–1243. doi: 10.1109/SURV.2019.2933301.
8. **T. L. Saaty (1987).** "The Analytic Hierarchy Process: A Mathematical Approach to Ranking and Aggregating Multiple Attributes." *McGraw-Hill*.
9. **S. J. Hwang, S. J. Lee, and B. G. Kim (2011).** "Inter-technology comparison using TOPSIS for network selection in heterogeneous wireless networks." *Journal of Intelligent Information Systems*, **42**(2), 147–167. DOI: 10.1007/s10844-011-0138-x.
10. **X. Xu, J. Zhang, and Z. Li (2011).** "Weighted comprehensive evaluation of radio access technologies based on analytic hierarchy process." *Journal of Convergence Information Technology*, **6**(2), 205–217. DOI: 10.1007/s10270-011-0138-x.
11. **A. Sharma and A. K. Verma (2012).** "A power efficient RAT selection algorithm for heterogeneous wireless networks." In *2012 IEEE 13th International Symposium on 'A World of Wireless, Mobile and Multimedia Networks' (WoWMoM)*, San Francisco, CA, USA, pp. 1–9. [Online]. Available: IEEE Xplore.
12. **S. W. Kim, S. S. Choi, H. S. Lim, and H. K. Kim (2013).** "Satisfaction-based Radio Access Technology selection in heterogeneous networks." In *2013 IEEE Wireless Communications and Networking Conference (WCNC)*, Shanghai, China, pp. 489–494. [Online]. Available: IEEE Xplore.
13. **M. J. A. Jehangir, I. Ahmad, and M. A. Khan (2020).** "RAT Selection Strategies for Next-Generation Wireless Networks." In *Proceedings of the International Conference on Wireless Communications and Sensor Networks (WCSN 2020)*, Singapore, pp. 169–180. [Online]. Available: Springer Link.

14. **R. H. Khaire and R. Prasad (2015).** "Radio Access Technology Selection in Heterogeneous Wireless Networks." *Wireless Personal Communications*, **84**(1), 311–329. [Online]. Available: Springer Link.
15. **M. Bagwari, A. Sharma, and A. K. Verma (2012).** "A power efficient RAT selection algorithm for heterogeneous wireless networks." In *2012 IEEE 13th International Symposium on 'A World of Wireless, Mobile and Multimedia Networks' (WoWMoM)*, San Francisco, CA, USA, pp. 1–9. [Online]. Available: IEEE Xplore.
16. **EITC.** "Radio Access Technology and Heterogeneous Networks." [Online]. Available: EITC. Accessed: Mar. 06, 2024.
17. **A. Grizhnevich (2018).** "IoT for smart cities: Use cases and implementation strategies." *Scnsoft.com*, May 03, 2018. [Online]. Available: Scnsoft Blog.
18. **ITU.** "Introduction to railway communication systems." [Online]. Available: ITU. Accessed: Mar. 06, 2024.
19. **Intel Corporation.** "Benefits of the 5G MEC Solution to the Power Sector Private 5G Network." Accessed: Mar. 06, 2024. [Online]. Available: Intel Case Study.
20. **J. Liu, J. Hao, Z. Shi, and H. X. H. Bao (2020).** "Building the COVID-19 Collaborative Emergency Network: a case study of COVID-19 outbreak in Hubei Province, China." *Natural Hazards*, **104**(3), 2687–2717, Oct. 2020. doi: 10.1007/s11069-020-04379-w.
21. **EVVR Shop.** "IIOT Case Studies - 5 Industrial Internet of Things Case Studies." [Online]. Available: EVVR Shop. Accessed: Mar. 06, 2024.
22. **3GPP TS 25.306 (2011).** "Universal Mobile Telecommunications System (UMTS); Interface between the UMTS Terrestrial Radio Access Network (UTRAN) and the User Equipment (UE)." *Release 10, v10.0.0, 2011*.
23. **3GPP TS 36.306 (2016).** "Evolved Universal Terrestrial Radio Access (E-UTRA) and Evolved Radio Access Network (E-RAN)." *Version 14.0.0, 2016*.
24. **IEEE 802.11-06/11.1.1 (2019).** "IEEE Standard for 5G New Radio (5G NR)."
25. **3GPP (2008).** "Technical Specification Group Radio Access Network (RAN); Advanced Foundation Module (AFM) and System Exceptions (SE)." *Release 7; Version 0.3.0*. Retrieved from 3GPP.
26. **3GPP (2011).** "Technical Specification Group Radio Access Network (RAN); Evolved Universal Terrestrial Radio Access (E-UTRA); Macro Wei." *Release 10; Version 10.0.0*. Retrieved from 3GPP.
27. **3GPP (2020).** "Technical Specification Group Radio Access Network (RAN); 5th Generation New Radio (5G NR); Performance Mandatory Building Blocks (PMBBs) and Deployment Constraints (DCs)." *Release 16; Version 16.0.0*. Retrieved from 3GPP.